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NEWS AND PERSPECTIVES

A first report of separation calls in southern yellow-cheeked gibbons (*Nomascus gabriellae*) in captivity

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Abstract
The effects of social separation, including vocalization, have been studied for a very long time in non-human primates under laboratory conditions. As part of the long-term research on the vocal behaviour of *Nomascus* gibbons in zoos, this study provides the first record of calls of the southern yellow-cheeked gibbon (*Nomascus gabriellae*) in response to involuntary separation. Our study revealed that calls were also emitted by an infant (aged 1 year 8 months), and that the acoustic structure of the infant's calls was similar to that of older individuals' calls. Separation-induced calls seem to have a shorter developmental convergence than vocalizations with a stable pattern (which are specific for species and sex). The acoustic structure of the calls reported here comprised simple syllables, and differed from the sex- and species-specific vocal patterns of this species. Our findings demonstrate a novel paradigm in this genus, and provide evidence of the ability of gibbons to express distress when socially separated.

Keywords Separation call · Vocalization · Gibbons · *Nomascus*

Introduction

In mammalian species, involuntary disruption of social bonds invariably stimulates a characteristic separation response, consisting of behavioural agitation, the tendency to seek reunion with group members and adrenal activation (Ainsworth 1972; Bowlby 1973). The effects of social separation, including vocalization, have been studied primarily in laboratory non-human primates (Hennessy 1997). Most of these studies have focused on the separation of infants from their mothers under laboratory conditions, e.g. rhesus macaque [*Macaca mulatta* (Bayart et al. 1990)], squirrel monkeys (*Saimiri* spp. (Masataka and Symmes 1986; Symmes and Biben 1985)], common marmoset [*Callithrix jacchus* (Yamaguchi et al. 2010), titi monkeys [*Callicebus* spp. (Cubicciotti and Mason 1976; Fernandez-Duque et al. 1997; Hoffman et al. 1995] and tufted capuchin [*Cebus apella* (Byrne and Suomi 1999)]. Only a few similar studies have been undertaken on non-human primates in the wild, e.g. rhesus macaque [*M. mulatta* (Berman et al. 1994)] and the chacma baboon [*Papio cynocephalus ursinus* (Rendall et al. 2000)]. With respect to other species of non-human primates, including gibbons (family Hylobatidae), vocalization during involuntary separation outside laboratory conditions has received little attention.

Among non-human primates, gibbons are a relatively small, uniform group of territorial and arboreal apes that mostly have a monogamous social structure. Along with other members of the ape superfamily (Hominoidea), they are among the closest extant relatives of humans (Roos and Geissmann 2001). Unlike great apes, however, all gibbon species are well known for emitting a loud and stable pattern of vocalizations, which are innate and specific for species and sex (Haimoff 1984). Although this stable vocal pattern, including that of male calls and the great call in females, has been relatively well studied in a number of gibbon species (Geissmann 2002), little is known about other vocalizations in gibbons. However, there are two exceptions to this. Firstly, it was determined that white-handed gibbons (*Hylobates lar*) are able to assemble syllables of their stable songs (sex- and species-specific vocal patterns) into alarm calls when

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they detect a potential predator (Clarke et al. 2006, 2015). Secondly, Hainan gibbons (*Nomascus hainanus*), with the exception of infants, are able to produce non-specific alarm calls when they detect a threat (birds of prey, snakes or humans) (Deng et al. 2016).

This study reports a new type of vocalization of the southern yellow-cheeked gibbon (*Nomascus gabriellae*), including that of an infant (less than 2 years of age), in response to involuntary separation. This research comprises part of a long-term study of the vocal behaviour of *Nomascus* gibbons in zoological parks. The calls were recorded from four individuals of various age categories in three different zoological parks. The aim of this study was to analyse the acoustic structure of calls emitted by captive southern yellow-cheeked gibbons upon involuntary social separation.

Methods

Study subjects

Four southern yellow-cheeked gibbons kept in zoos in the Czech Republic (Olomouc and Jihlava) and the Slovak Republic (Bojnice) were included in this study (Table 1). The gibbons were kept in family groups consisting of an adult male and female and their offspring. In accordance with Ellefson (1974), the monitored individuals were categorized as infants (~0–2 years of age), immatures (~2–6 years of age), sub-adults (~6–8 years of age) or adults (8+ years of age). Two of the zoos (Olomouc and Jihlava) held two groups of gibbons that remained in visual and auditory contact with each other. The indoor and outdoor enclosures were equipped with platforms at various heights, trees and extensive rope systems. The gibbons were fed four times a day and water was available ad libitum. The diet consisted of fruit and vegetables, seeds, leaves, cereals and an occasional egg.

Data collection and acoustic analysis

The recordings were made under ordinary circumstances. Each group of gibbons had permanent access to an indoor and outdoor enclosure. Regular cleaning of the outdoor enclosure meant that the gibbons were fed in an indoor enclosure, which the animals entered via a trap door after being lured with food. The keeper closed the trap door quickly after the gibbons had entered the indoor enclosure to ensure that they did not return outdoors. However, sometimes the keeper failed to close the door in time, leading to the inadvertent separation of an individual gibbon from its group. Immediately after such an event, the isolated gibbon uttered a call. The duration of the separation lasted until the keeper reopened the trap door and the gibbon was able to re-enter the indoor enclosure. We recorded the calls of the separated individuals in Waveform Audio File Format on an M-Audio Micro Track II recorder or Marantz PMD661 with a Rode NTG-2 semi-directional microphone (16-bit resolution, 44.1-kHz sampling rate). The recordings were made within a distance of 2 to 4 m from the animals. For analysis, the sampling frequency was reduced from 44.1 kHz to 12 kHz. Spectrograms were generated using Avisoft–SASLab Pro version 5.2 software (Berlin) at the following settings: fast Fourier transformation length = 1024, frequency resolution = 12 Hz, temporal resolution = 21.3 ms, overlap = 75%, window type = Hamming. First inspection of data showed that the patterns of the separation calls differed from the common loud and stable patterns of vocalizations produced by gibbons (Konrad and Geissmann 2006).

In order to analyze the vocalizations, which consisted of single syllables, a basic spectro-temporal acoustic analysis

Table 1 Subject characteristics and information regarding family group composition

General information				
Zoo	Bojnice	Jihlava		Olomouc
No. of the subject, place and date of birth as recorded in the *European studbook* (Lefaux 2013)	No. 148, Bojnice, 2 April 2008	No. 173, Jihlava, 2 April 2011	No. 213, Jihlava, 25 November 2015	No. 54, in the wild, ~1992
Date, age class and age at the time of recording the calls	16 October 2014, sub-adult male, 6 years 6 months	20 May 2015, immature female, 4 years 1 month	6 July 2017, infant male, 1 year 8 months	10 October 2014, adult female, ~20 years
Information about family groups (sex, no.) given in the *European studbook* (Lefaux 2013), and age classes as defined by Ellefson (1974)	Adults: Adult ♂, no. 37 Adult ♀, no. 80 Offspring: Sub-adult ♂, no. 148 Immature ♀, no. 162 Infant ♂, no. 184	Adults: Adult ♂, no. 36 Adult ♀, no. 84 Offspring: Immature ♂, no. 173 Infant/immature ♀, no. 192 Infant ♂, no. 213		Adults: Adult ♂, no. 113 Adult ♀, no. 54 Offspring: Immature ♂, no. 178 Infant ♂, no. 194

was applied for each syllable (Fig. 1). We measured the onset frequency (the starting frequency of each syllable), the highest frequency (the maximum frequency of each syllable), the offset frequency (the ending frequency of each syllable) and the duration of the syllables (the time between the beginning and end of each syllable).

Statistical analysis

The dataset consisted of records covering 167 syllables uttered by four individuals of different age and sex (Table 1). These single syllables were characterized by four variables: the onset frequency, offset frequency, highest frequency and the duration. Due to the small samples size ($n = 4$), only descriptive statistics are expressed, as the mean ± SD, for every variable and individual.

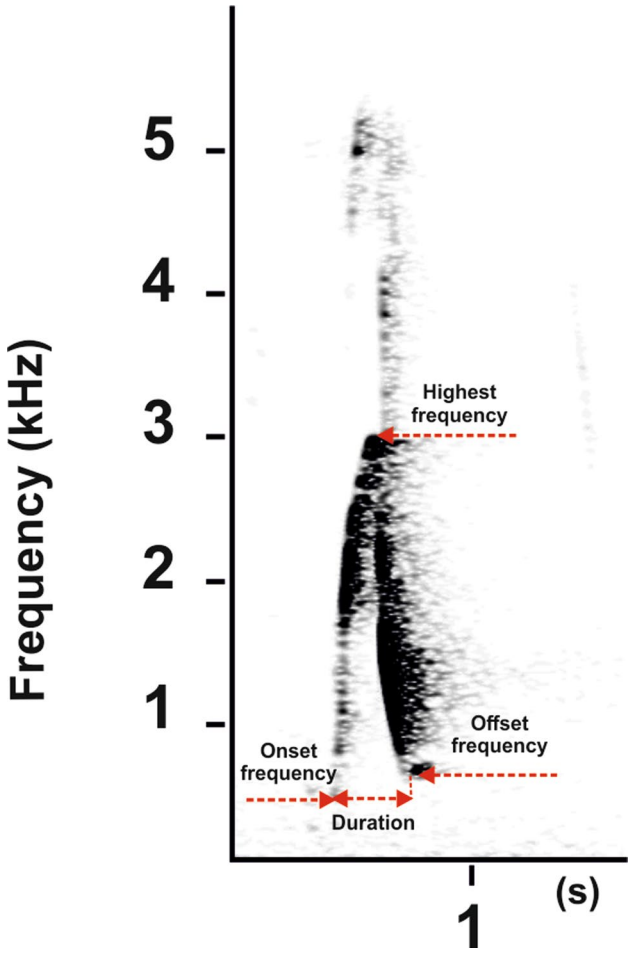


Fig. 1 A representative spectrogram indicating the measured parameters of a call syllable: onset frequency (the starting frequency of each syllable), highest frequency (the maximum frequency of each syllable), offset frequency (the end frequency of each syllable) and the duration of a syllable (the time between the start and end of each syllable)

Results

Calls were emitted by all individuals almost immediately upon separation from the group when they were left alone in the outdoor enclosure. The latency between the initial separation and the commencement of vocalization ranged from 2 to 4 s (sub-adult male, 2 s; immature female, 3 s; infant male, 4 s; and adult female, 3 s). All of the separated gibbons ceased their calls once the trap door had been opened and they were able to rejoin their groups. All of the calls consisted of simple syllables (Fig. 2). The acoustic parameters for each syllable are presented in Table 2.

Discussion

Our study provides evidence that captive southern yellow-cheeked gibbons emit calls due to involuntary social separation. The acoustic structure of the separation-induced calls comprised only simple syllables, in contrast to the remarkably complex sex- and species-specific vocal patterns previously recorded in this species (Geissmann 2002). These unexpected vocalizations were recorded by chance following the separation of four animals of different age categories in three different zoological parks. This suggests that these vocalizations are of general importance, despite the low number of gibbons involved in the study.

Furthermore, our study also revealed that an infant of less than 2 years of age (1 year 8 months) uttered a call that had the same pattern as those of the older separated individuals (e.g. simple syllables of the same duration and the same highest onset and offset frequencies). It takes several years for gibbon offspring to attain the sex- and species-specific vocal patterns of their parents (Hradec et al. 2016; 2017; Koda et al. 2013; 2014; Merker and Cox 1999). Therefore, there does not seem to be a slow developmental pathway for separation-induced calls common to both parents and offspring, as is the case for the well-known stable, sex-specific and species-specific vocal patterns of adult gibbons.

In the present study, the four gibbons most likely found social separation highly stressful, which they indicated by immediately calling out to their group members. This behaviour continued until the separated gibbon was reunited with its group, despite the fact that all of the animals were acquainted with the outdoor enclosure. However, the mean call rate per minute differed between individual gibbons, which suggests that their perception of separation differed. Only further research can determine whether call rate is influenced by age and/or individuality.

Non-human primate species react differently to total separation or partial separation from their mother or group

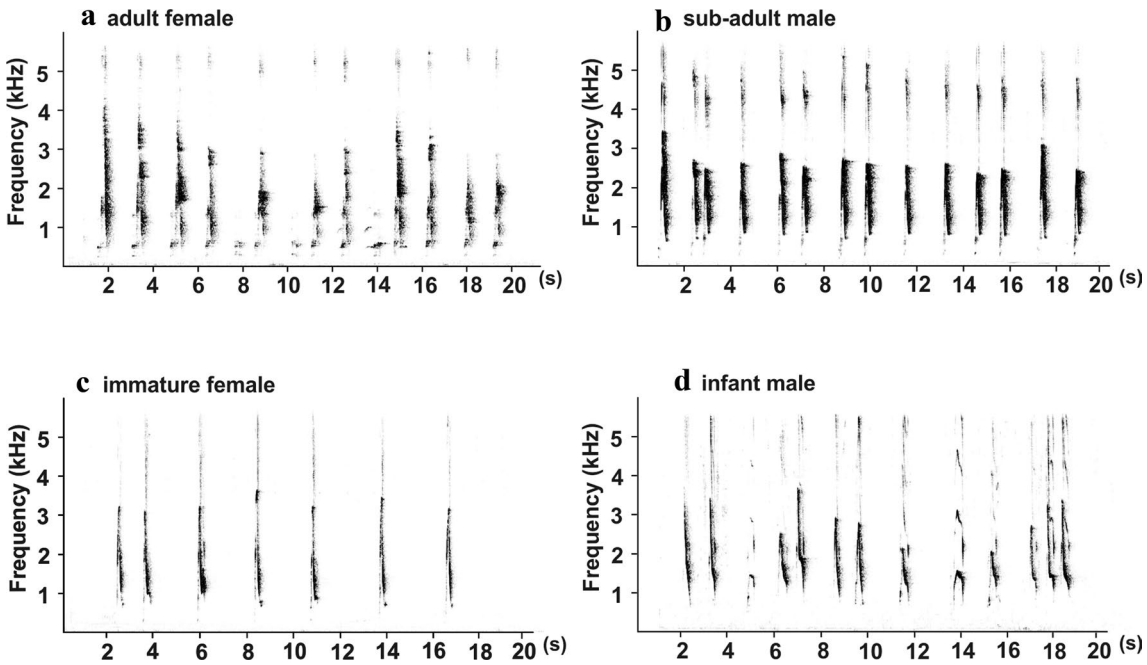


Fig. 2 Representative spectrograms of calls emitted by the gibbons upon involuntary separation from their groups

Table 2 Acoustic parameters of separation-induced calls (mean ± SD) in southern yellow-cheeked gibbons

Subjects	Number of syllables	Length of time the calls were voiced for (min:s)	Rate of syllables per minute (mean ± SD)	Syllable duration (s) (mean ± SD)	Onset frequency (Hz) (mean ± SD)	Highest frequency (Hz) (mean ± SD)	Offset frequency (Hz) (mean ± SD)
Sub-adult male	66	1:44	33 ± 6	0.23 ± 0.06	290 ± 138	2610 ± 226	730 ± 73
Immature female	49	5:59	8.17 ± 3.76	0.34 ± 0.07	300 ± 164	3190 ± 343	670 ± 99
Male infant	34	11:27	2.83 ± 3.91	0.33 ± 0.06	355 ± 133	2830 ± 687	970 ± 207
Adult female	18	2:18	6 ± 5.1	0.21 ± 0.06	470 ± 94	3200 ± 184	680 ± 179
Mean			12.5 ± 4.69	0.28 ± 0.06	354 ± 132	2958 ± 360	763 ± 140

members under laboratory conditions (Bayart et al. 1990; Byrne and Suomi 1999; Yamaguchi et al. 2010). Rhesus macaques emitted fewer calls in response to total separation compared to when they had visual and/or auditory contact with their mothers or members of their family (Bayart et al. 1990; Kalin et al. 1992). This is in contrast to common marmosets, titi monkeys and tufted capuchins, which emitted more calls in response to total separation (Byrne and Suomi 1999; Cubicciotti and Mason 1976; Hoffman et al. 1995; Yamaguchi et al. 2010).

The question remains whether gibbons in the wild emit separation-induced calls of similar acoustic structure when they lose contact with group members. There is some evidence that certain non-human primate species use the same type of vocal structure upon involuntary separation, whether in the laboratory or in the wild [e.g. phee calls of common marmosets (Epple 1968; Norcross and Newman 1993; Stevenson and Rylands 1988; Yamaguchi et al. 2010) and peep calls of squirrel monkeys (Boinski and Newman 1988; Winter et al. 1966; Winter 1972)].

As mentioned in the Introduction, there is little information available on other vocalizations (e.g. alarm calls) in any of the gibbon species. Surprisingly, the spectrograms analysed in this study showed the following similarities with the second part of the non-specific alarm calls (i.e. mobbing calls) of Hainan gibbons (Deng et al. 2016): identical structure, frequency range of vocalization (190–3930 Hz this study vs. 690–3920 Hz mobbing calls) and nearly identical maximum frequencies (2958 ± 360 Hz this study vs. 2970 ± 600 Hz mobbing calls). Speculatively, Hainan and southern yellow-cheeked gibbons, which are both members of the genus *Nomascus*, may share at least some characteristics of their non-specific distress vocalizations emitted in response to different threats (Deng et al. 2016). Future

experiments should help to clarify whether southern yellow-cheeked gibbons in captivity and in the wild use the same calls in the presence of threats (e.g. predators) or upon separation from their social group.

In conclusion, our study is the first to provide evidence that southern yellow-cheeked gibbons of different age categories, including infants, emit calls in response to involuntary separation from their social groups.

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Compliance with ethical standards

Conflict of interest The authors declare that they have no conflict of interest.

Ethics All applicable international and institutional guidelines for the care and use of animals were followed. The zoological institutions in which the animals in this study live maintain rigorous standards of animal welfare and are accredited by the European Association of Zoos and Aquaria and the Union of Czech and Slovak Zoos. This study was approved by the Ethics and Animal Care Committee of the Czech University of Life Sciences Prague (protocol no. CZU/1606) and was fully non-invasive. This study fully complied with the legal requirements of the Czech Republic and the Slovak Republic, as well as those stipulated by European Directive 2010/63/EU.

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